TUTORIAL 3

Linear Systems and Subspaces

MOTIVATION AND GUIDING IDEA

Row reduction replaces a problem with an equivalent one that is easier to read. After calculating, return to the question: is the system consistent, which variables are free, and what is the solution set? For a nonhomogeneous system, the difference of two solutions solves the associated homogeneous system. This explains the form ``a particular solution plus a homogeneous solution.''

NOTATION AND CONVENTIONS

If $A \in M_{m\times n}(F)$, then $Ax = b$ is a system of m equations in n variables, with $x \in F^n$ and $b \in F^m$. The notation $[A \mid b]$ denotes its augmented matrix, and $A \sim B$ denotes row equivalence. The rank $\operatorname{rank} A$ is the number of pivots. Define $\ker A = \{x \in F^n : Ax = 0\}$. The expression $\operatorname{span}\{v_1, \dots, v_k\}$ denotes all combinations $\alpha_1 v_1 + \cdots + \alpha_k v_k$ with $\alpha_j \in F$. The notation $x_0 + U = \{x_0 + u : u \in U\}$ denotes a translate of a subspace U. Distinguish the empty set $\varnothing$, with no elements, from $\{0\}$, which has one element.

DEFINITION 3.1

Vector space axioms and solution tools. A vector space over a field F is a set V equipped with operations $V \times V \to V$ and $F \times V \to V$ satisfying, for all $u, v, w \in V$ and $\alpha, \beta \in F$,

$$u + v = v + u, \qquad (u + v) + w = u + (v + w).$$

There is $0_V \in V$ such that $v + 0_V = v$, and every $v \in V$ has an additive inverse $-v \in V$ satisfying $v + (-v) = 0_V$. In addition,

$$\alpha(u + v) = \alpha u + \alpha v, \qquad (\alpha + \beta)v = \alpha v + \beta v,$$

$$(\alpha\beta)v = \alpha(\beta v), \qquad 1_F v = v.$$

For $A \in M_{m\times n}(F)$, the system $Ax = b$ is consistent if and only if $\operatorname{rank} A = \operatorname{rank}[A \mid b]$. If x_0 is a particular solution, then its full solution set is $x_0 + \ker A$: indeed, $Ax = b$ is equivalent to $A(x - x_0) = 0$. This solution set is a subspace of F^n if and only if $b = 0$, since containing the zero vector forces $A0 = b = 0$.

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MOTIVATION AND GUIDING IDEA

Objectives. We will formulate problems as linear systems, perform row reduction while preserving the solution set, and distinguish free variables from pivot variables. We will connect homogeneous systems with vector spaces and test closure systematically.

3.1 Linear systems and row reduction

3.1.1 A system of linear equations

RECALL AND TOOLS

In general, for a positive integer n, a linear equation in n unknowns (variables) has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

Here $a_1, a_2, ..., a_n$ and b are **given scalars** in a specified field $\mathbb{F}$, such as $\mathbb{R}$, $\mathbb{C}$, or a finite field, and $x_1, x_2, ..., x_n$ are the **unknowns**, or variables.
The scalar a_i is the **coefficient** of x_i, and b is the **constant term** of the equation.
A system of m equations in n unknowns has the general form

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n = b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n = b_2 \\ \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n = b_m \end{cases}$$

To solve the system, it is convenient to collect the **coefficients** in a matrix, preserving the variable order. To shorten the notation, we write either the matrix of variable coefficients, called the **coefficient matrix**, or the **augmented matrix**, which also includes the constant terms. Following the basic rules consistently makes it clear which coefficient belongs to each variable.

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EXAMPLE 3.1

$$\begin{cases} 3x_1 - 2x_2 = 0, \\ -6x_1 + 5x_2 = 2 \end{cases} \longleftrightarrow \left(\begin{array}{cc|c} 3 & -2 & 0 \\ -6 & 5 & 2 \end{array}\right).$$

3.1.2 Elementary operations

DEFINITION 3.2

Elementary row operations:
An elementary row operation modifies a matrix in one of the following ways:

1. Interchange rows i and j, denoted by $R_i \longleftrightarrow R_j$.
2. Multiply row i by a **nonzero** scalar λ, denoted by $R_i \longleftarrow \lambda R_i$.
   (Dividing by a nonzero scalar a is multiplication by $1/a$.)

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3. Add λ times row j to a different row i, where $i \neq j$, denoted by $R_i \longleftarrow R_i + \lambda R_j$.

- Elementary row operations preserve the solutions of a linear system.
- Every elementary row operation has an inverse that is also an elementary row operation.

DEFINITION 3.3

Matrices A and B are **row equivalent** (or simply equivalent), denoted by $A \sim B$, if B is obtained by applying a finite sequence of elementary row operations to A.

DEFINITION 3.4

Leading entry (pivot): in every nonzero row of a matrix, the first nonzero entry is called its leading entry.

DEFINITION 3.5

Row echelon form: each leading entry is strictly to the left of the leading entry in the next nonzero row. Any zero rows appear below all nonzero rows.

DEFINITION 3.6

Reduced row echelon form: a matrix in row echelon form such that, in each nonzero row, the leading entry equals 1, and every other entry above and below it in the same column equals 0. These requirements apply to every column, including the constant column of an augmented matrix.

PROPOSITION 3.1

Every matrix is row equivalent to a unique matrix in reduced row echelon form.

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Examples:

1. $\begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$ is not in echelon form: its leading entries are in the same column.
2. $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{pmatrix}$ is in echelon form but not reduced echelon form.
3. $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \end{pmatrix}$ is in reduced row echelon form.

3.1.3 Gaussian elimination

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RECALL AND TOOLS

1. Start with the first active row and the first column. Look in that column for a nonzero entry in the active row or below it. If none exists, move to the next column.
2. Interchange rows to bring this entry into the active row. Division by it is permitted because it has been checked to be nonzero.
3. Multiply the active row by the inverse of its leading entry to obtain a leading 1, and use it to eliminate the entries below.
4. Move to the next active row and seek a pivot column to the right of the previous one. A pivot column need not have the same index as its row.
5. After reaching row echelon form, eliminate the entries above each pivot, working from the last pivot to the first, to obtain reduced row echelon form. This completes Gauss-Jordan elimination.

EXERCISE 3.1

Please reduce the augmented matrix over $\mathbb{R}$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 1 & 3 & 12 \\ 1 & 3 & 2 & 3 \end{array}\right)$$

to reduced row echelon form and please find the system's solution.

WORKED SOLUTION

Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -3 & 1 & 10 \\ 0 & 1 & 1 & 2 \end{array}\right).$$

Swap the last two rows and apply $R_3 \leftarrow R_3 + 3R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 4 & 16 \end{array}\right).$$

Divide R_3 by 4, subtract it from R_2, and clear the entries above the last two pivots:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{array}\right).$$

The unique solution is $(1, -2, 4)^T$. Substitution gives $1-4+4 = 1$, $2-2+12 = 12$, $1-6+8 = 3$.

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NOTE

To solve a linear system, reduce its **augmented matrix**, including the constant terms. You may stop at row echelon form and use back substitution, or continue to reduced row echelon form and read off the solutions.

EXAMPLE 3.2

Consider the system

$$\begin{cases} x + 2y = 0, \\ y + z = 2, \\ 2x + z = 7. \end{cases}$$

Its augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 1 & 7 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 5 & 15 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right).$$

The first step uses $R_3 \leftarrow R_3 - 2R_1 + 4R_2$. Hence $z = 3$, $y = 2 - z = -1$ and $x = -2y = 2$. The solution set is $\{(2, -1, 3)\}$.

This example shows how row reduction helps solve a linear system. We will next see that a system need not have a unique solution.

EXERCISE 3.2

Please reduce the augmented matrix over $\mathbb{Z}_{11}$

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 & 4 \\ 2 & 3 & 4 & 1 & 5 \end{array}\right)$$

to reduced row echelon form. Please perform the arithmetic modulo 11.

WORKED SOLUTION

Apply $R_3 \leftarrow R_3 - 2R_1 - 3R_2$. The last row has entries $4 - 4 - 9 = -9 \equiv 2$, $1 - 2 - 6 = -7 \equiv 4$ and $5 - 6 - 12 = -13 \equiv 9$. Thus

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 & 4 \\ 0 & 0 & 2 & 4 & 9 \end{array}\right).$$

The inverse of 2 is 6. Multiplying R_3 by 6 gives

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 & 4 \\ 0 & 0 & 1 & 2 & 10 \end{array}\right).$$

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EXERCISE 3.2 CONTINUED

Finally apply $R_1 \leftarrow R_1 - 2R_3$ and $R_2 \leftarrow R_2 - 3R_3$:

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 8 & 5 \\ 0 & 1 & 0 & 7 & 7 \\ 0 & 0 & 1 & 2 & 10 \end{array}\right).$$

All entries use representatives $0, 1, \ldots, 10$.

EXERCISE 3.3

Please solve the following system over $\mathbb{C}$:

$$\begin{cases} iz + (2-i)w = -3 - i, \\ 2iz + (5-2i)w = -5 - 5i. \end{cases}$$

Please verify the solution in both equations.

WORKED SOLUTION

Subtract twice the first equation from the second to get

$$w = (-5 - 5i) - 2(-3 - i) = 1 - 3i.$$

Now $(2-i)(1-3i) = -1 - 7i$, so

$$iz = -3 - i - (-1 - 7i) = -2 + 6i, \qquad z = 6 + 2i.$$

The first equation gives $(-2 + 6i) + (-1 - 7i) = -3 - i$. The second gives

$$2iz + (5 - 2i)w = (-4 + 12i) + (-1 - 17i) = -5 - 5i.$$

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3.2 Solving linear systems

DEFINITION 3.7

After row reduction, a variable whose column in the coefficient matrix contains a leading entry is called a **pivot variable** (or dependent variable). A variable whose column contains no leading entry is called a **free variable**. In a consistent system, every free variable can be assigned an arbitrary value in the field, after which the pivot variables are determined by the equations. Over an infinite field, a free parameter gives infinitely many choices; over a finite field, the number of choices is finite. The number of free variables is the number of degrees of freedom of a consistent system.

DEFINITION 3.8

The rank of a matrix A, denoted by $\operatorname{rank}(A)$, is the number of nonzero rows in a row echelon form. Equivalently, it is the number of leading entries (pivots) in that form.

Remark: if a system with coefficient matrix A and n variables is consistent, its number of degrees

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of freedom is $n - \operatorname{rank}(A)$. Over $\mathbb{F}_q$, it has exactly $q^{n-\operatorname{rank}(A)}$ solutions.

EXERCISE 3.4

Please solve over $\mathbb{R}$ the system

$$\begin{cases} x + 3y - 2z = 7, \\ 2x - y + 3z = 0, \\ 4x + 5y - z = 14. \end{cases}$$

WORKED SOLUTION

Write the augmented matrix and subtract multiples of its first row:

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & 7 \\ 2 & -1 & 3 & 0 \\ 4 & 5 & -1 & 14 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 3 & -2 & 7 \\ 0 & -7 & 7 & -14 \\ 0 & -7 & 7 & -14 \end{array}\right).$$

Subtract row two from row three, divide row two by -7 and clear its pivot column:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

Put $z = t$. Then $x = 1 - t$, $y = 2 + t$, so

$$S = (1, 2, 0) + \operatorname{span}\{(-1, 1, 1)\}.$$

There is one free variable and infinitely many solutions. Substitution gives $7, 0, 14$ in the three equations for every $t \in \mathbb{R}$.

EXERCISE 3.5

Please solve the system

$$\begin{cases} 2x_1 - 6x_2 + 4x_3 = 7, \\ x_1 - 3x_2 + 2x_3 = 2. \end{cases}$$

WORKED SOLUTION

Swap the rows, then subtract twice row one from row two:

$$\left(\begin{array}{ccc|c} 2 & -6 & 4 & 7 \\ 1 & -3 & 2 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -3 & 2 & 2 \\ 0 & 0 & 0 & 3 \end{array}\right).$$

The last row says $0 = 3$, so there is no solution.

Geometric illustration

We have encountered three possible cases:

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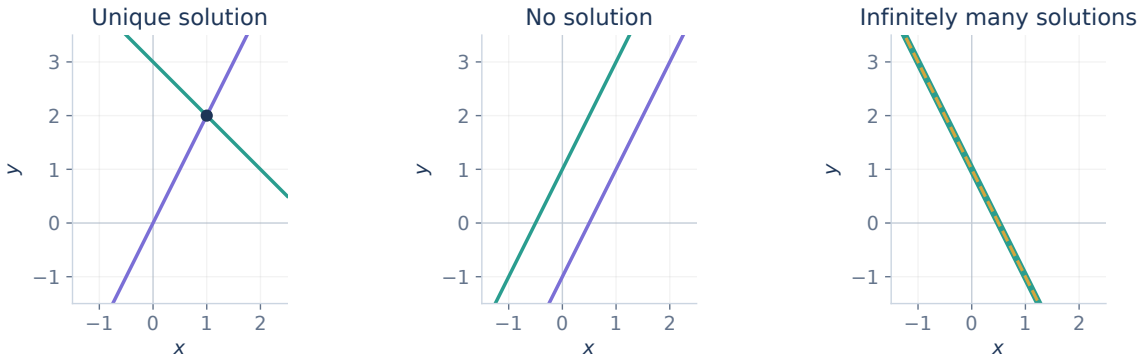


Figure 3.1. Over the real numbers, two equations in two unknowns can describe intersecting lines, distinct parallel lines, or coincident lines. These give, respectively, a unique solution, no solution, or infinitely many solutions.

EXERCISE 3.6

Let $a \in \mathbb{R}$. Please classify the number of solutions and please state the coefficient matrix rank in each case:

$$\begin{cases} x - 2y = 0, \\ 3x + 3z = 2, \\ x + az = 1. \end{cases}$$

WORKED SOLUTION

Subtract three times row one from row two, and row one from row three. Swap the last two rows and subtract three times row two from row three:

$$\left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 3 & 0 & 3 & 2 \\ 1 & 0 & a & 1 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 2 & a & 1 \\ 0 & 0 & 3-3a & -1 \end{array}\right).$$

If $a = 1$, there is a contradiction row and rank 2. If $a \neq 1$, the rank is 3 and the solution is unique:

$$z = -\frac{1}{3-3a}, \qquad x = 1 + \frac{a}{3-3a}, \qquad y = \frac{1}{2}\left(1 + \frac{a}{3-3a}\right).$$

No value of a gives infinitely many solutions.

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3.2.1 Homogeneous systems

RECALL AND TOOLS

All constant terms are 0. In this case:

- We may write only the coefficient matrix rather than the augmented matrix: the zero column is unchanged by all row operations, so there is no need to keep writing it.
- A solution always exists: no contradiction row can occur, so inconsistency is impossible. We must still determine whether the zero solution is the only solution. Over an infinite field, the existence of a nonzero solution implies infinitely many solutions.
- If there is a unique solution, it is the zero solution, also called the **trivial solution**.

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- The solution set of a homogeneous system in n variables is a vector subspace of $\mathbb{F}^n$ and can always be expressed as a span.

EXERCISE 3.7

Let $a \in \mathbb{R}$. Please classify the number of solutions of this homogeneous system and please find the solution set in each case:

$$\begin{cases} x - 2y = 0, \\ 3x + 3z = 0, \\ x + az = 0. \end{cases}$$

WORKED SOLUTION

The coefficient matrix is the same as in the preceding exercise. Row reduction gives

$$\begin{pmatrix} 1 & -2 & 0 \\ 3 & 0 & 3 \\ 1 & 0 & a \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 0 \\ 0 & 2 & a \\ 0 & 0 & 3-3a \end{pmatrix}.$$

If $a \neq 1$, the rank is 3 and only the zero solution exists. If $a = 1$, the rank is 2, and $x = -z$, $y = -z/2$. Choosing $z = 2t$ gives

$$S = \{(-2t, -t, 2t) : t \in \mathbb{R}\} = \text{span}\{(-2, -1, 2)\}.$$

There are infinitely many solutions in this case. A homogeneous system is always consistent, so no case has zero solutions.

EXERCISE 3.8 EXAM QUESTION

Please work in $\mathbb{Z}_7$. Please solve the system in terms of $\omega \in \mathbb{Z}_7$:

$$\begin{cases} x + y + z = 2, \\ x + \omega y + z = 5, \\ x + y + (\omega + 1)z = 2. \end{cases}$$

Please write the solution set and number of solutions in every case.

WORKED SOLUTION

Subtract the first equation from each of the other two:

$$\begin{cases} x + y + z = 2, \\ (\omega - 1)y = 3, \\ \omega z = 0. \end{cases}$$

1. If $\omega = 1$, the equation $0 = 3$ gives no solution.
2. If $\omega = 0$, then $-y = 3$, so $y = 4$. Choosing $z = t$ gives

$$S_0 = \{(5 - t, 4, t) : t \in \mathbb{Z}_7\}.$$

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EXERCISE 3.8 EXAM QUESTION CONTINUED

There are exactly seven solutions, since different values of t give different third coordinates.

3. If $\omega \in \{2,3,4,5,6\}$, both ω and $\omega - 1$ are invertible. Thus

$$z = 0, \qquad y = 3(\omega - 1)^{-1}, \qquad x = 2 - y.$$

Each case has a unique solution:

ω	2	3	4	5	6
x	6	4	1	3	0
y	3	5	1	6	2
z	0	0	0	0	0

As a check, the coefficient determinant is $\omega(\omega - 1)$, which vanishes only at $\omega = 0, 1$.

3.3 Vector spaces

RECALL AND TOOLS

1. To define a vector space V over a field $\mathbb{F}$, we specify an addition of vectors in V and a multiplication of a vector in V by a scalar in $\mathbb{F}$, satisfying the vector space axioms. These are **in addition** to addition and multiplication in $\mathbb{F}$. A product of two vectors is not part of the vector space axioms; if needed later, it will be defined separately.
2. Every field is a vector space over itself, with the field's addition and multiplication.
3. A vector space V is always defined over a field $\mathbb{F}$. Always specify the field of scalars. For example, as discussed in the lecture, $\mathbb{C}$ can be regarded as a vector space over $\mathbb{C}$ or over $\mathbb{R}$. These are two different vector spaces.

EXAMPLE 3.3

Let $\mathbb{F}$ be a field.

(a) $\mathbb{F}^n$: the set of n-tuples over $\mathbb{F}$, where $n \in \mathbb{N}$.

(b) $M_{m\times n}(\mathbb{F})$: the set of $m \times n$ matrices over $\mathbb{F}$.

(c) $\mathbb{F}[X]$: the set of all polynomials with coefficients in $\mathbb{F}$.

EXERCISE 3.9

Please determine whether the following sets are subspaces of $\mathbb{R}[X]$ over $\mathbb{R}$:

1. $V = \{p(X) \in \mathbb{R}[X] : p(-2) = 0,\ p(3) \neq 0\}$.
2. $V = \{p(X) \in \mathbb{R}[X] : \exists c \in \mathbb{R} \setminus \{0\},\ p(c) = 0\}$.

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EXERCISE 3.9 CONTINUED

WORKED SOLUTION

1. No. The zero polynomial is absent. Addition also fails: for

$$p(X) = (X+2)(X-8), \qquad q(X) = (X+2)^2,$$

we have $p(-2) = q(-2) = 0$ and $p(3) = -25$, $q(3) = 25$, but

$$(p+q)(X) = (X+2)(2X-6), \qquad (p+q)(3) = 0.$$

2. No. The polynomials $(X-2)^2$ and $(X+2)^2$ belong to the set, with roots $2, -2$, respectively. Their sum is $2X^2 + 8 > 0$ for all real X, so it does not belong.

3.4 Subspaces

DEFINITION 3.9

Let V be a vector space over a field $\mathbb{F}$, and let $U \subseteq V$ be a subset that is itself a vector space over $\mathbb{F}$ under the addition and scalar multiplication inherited from V. Then U is a vector subspace of V.

PROPOSITION 3.2

Let V be a vector space over a field $\mathbb{F}$, and let $U \subseteq V$ be a subset. Then U is a vector subspace if and only if all of the following conditions hold:

1. $0 \in U$, where this is the zero vector of V.
2. U is closed under addition.
3. U is closed under scalar multiplication.

EXAMPLE 3.4

1. The set $V = \{p(X) \in F[X] : p(-3) = 0\}$ is a subspace of $F[X]$. It contains the zero polynomial. For $p, q \in V$ and $\alpha \in F$,

$$(p+q)(-3) = p(-3) + q(-3) = 0, \qquad (\alpha p)(-3) = \alpha p(-3) = 0.$$

Thus it is closed under addition and scalar multiplication.

2. For $n \geq 3$, the set $W = \{x \in \mathbb{R}^n : x_1 + x_3 = 0\}$ is a subspace. It contains zero. If $x, y \in W$, then

$$(x_1 + y_1) + (x_3 + y_3) = (x_1 + x_3) + (y_1 + y_3) = 0.$$

Also $(\alpha x)_1 + (\alpha x)_3 = \alpha(x_1 + x_3) = 0$ for every $\alpha \in \mathbb{R}$.

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EXERCISE 3.10

Please prove or please disprove:

1. The set $U = \{(a, 3a^2) : a \in \mathbb{R}\}$ is a subspace of $\mathbb{R}^2$.
2. The symmetric matrices form a subspace of $M_n(F)$.

WORKED SOLUTION

1. This is not a subspace: $(1, 3) \in U$, but
$$(1, 3) + (1, 3) = (2, 6) \notin U,$$
because a first coordinate of 2 requires second coordinate $3 \cdot 2^2 = 12$.
2. The zero matrix is symmetric. If $A^T = A$, $B^T = B$ and $\lambda \in F$, then
$$(A + B)^T = A^T + B^T = A + B, \qquad (\lambda A)^T = \lambda A^T = \lambda A.$$
Thus all three subspace-test conditions hold.

EXERCISE 3.11

Let $p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3$ be a real polynomial.

1. Please find all polynomials satisfying $p(0) = 0$, $p(2) = 4$, $p'(0) = 0$, $p'(1) = 0$. Please determine whether the solution set is a subspace and whether $t = 1$ is a local maximum or minimum.
2. Please repeat with $p(0) = 2$, $p(2) = 6$, $p'(0) = 0$, $p'(1) = 0$.
3. Add $p'(2) = 5$ to the second part. Please determine whether the system is consistent and whether its solution set is a subspace.

WORKED SOLUTION

We have $p'(t) = a_1 + 2a_2 t + 3a_3 t^2$. In part one, $a_0 = a_1 = 0$, while the remaining conditions give
$$4a_2 + 8a_3 = 4, \qquad 2a_2 + 3a_3 = 0.$$
Subtracting twice the second equation from the first gives $2a_3 = 4$, so $a_3 = 2$ and $a_2 = -3$. The unique solution is
$$p_1(t) = -3t^2 + 2t^3.$$
Check: $p_1(2) = -12 + 16 = 4$ and $p_1'(1) = -6 + 6 = 0$. The zero polynomial fails $p(2) = 4$, so the solution set is not a subspace. Also $p_1''(t) = -6 + 12t$, hence $p_1''(1) = 6 > 0$, giving a local minimum.
In part two, $a_0 = 2$, $a_1 = 0$, and the other equations are unchanged. Thus
$$p_2(t) = 2 - 3t^2 + 2t^3.$$
Again the solution is unique, its set is not a subspace, and $t = 1$ is a local minimum. In part

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EXERCISE 3.11 CONTINUED

three, this unique polynomial gives

$$p_2'(2) = -12 + 24 = 12 \neq 5.$$

The system is inconsistent, and the empty solution set is not a subspace.

3.5 Supplementary exercises: measurements and counting solutions

EXERCISE 3.12

Consider the system

$$x_1 + x_2 = 7, \qquad x_2 + x_3 = 11, \qquad x_1 + x_3 = 8.$$

Please find x_1, x_2, x_3. Please find the only value of c for which adding $x_1 + x_2 + x_3 = c$ leaves the solution set unchanged. Please determine what happens for $c = 14$.

WORKED SOLUTION

Add the first two equations and subtract the third:

$$2x_2 = 7 + 11 - 8 = 10, \qquad x_2 = 5.$$

Then $x_1 = 7 - 5 = 2$ and $x_3 = 11 - 5 = 6$. The third equation checks as $2 + 6 = 8$. Adding all three equations gives

$$2(x_1 + x_2 + x_3) = 7 + 11 + 8 = 26.$$

Thus the required value is $c = 13$. Setting $c = 14$ creates a contradiction, so the augmented system has no solution.

EXERCISE 3.13

Please solve over $\mathbb{Z}_3$ the system

$$\begin{cases} x + y + z = 1, \\ x + z = 2. \end{cases}$$

Please count its solutions and please compare with the same equations over $\mathbb{R}$.

WORKED SOLUTION

Subtracting the second equation from the first gives $y = -1 = 2$ in $\mathbb{Z}_3$. Choosing $z = t$ gives

$$(x, y, z) = (2 - t, 2, t), \qquad t \in \mathbb{Z}_3.$$

The three solutions are $(2, 2, 0), (1, 2, 1), (0, 2, 2)$. Over $\mathbb{R}$ we get

$$(x, y, z) = (2 - t, -1, t), \qquad t \in \mathbb{R},$$

so there are infinitely many solutions. Both systems have one free variable, but the number of solutions depends on the field size.

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Additional practice

EXERCISE 3.14 FOUNDATION

Please solve over $\mathbb{R}$ the system

$$\begin{cases} x + y + z = 6, \\ x - y + z = 2. \end{cases}$$

Please write its solution set as a particular solution plus multiples of a fixed vector. Please determine whether it is a subspace and please describe all differences between pairs of solutions.

WORKED SOLUTION

Subtracting the second equation from the first gives $2y = 4$, so $y = 2$ and $x + z = 4$. Put $z = t$ to get

$$S = \{(4 - t, 2, t) : t \in \mathbb{R}\} = (4, 2, 0) + \operatorname{span}\{(-1, 0, 1)\}.$$

It is not a subspace, because zero does not satisfy the equations. The difference of two solutions is

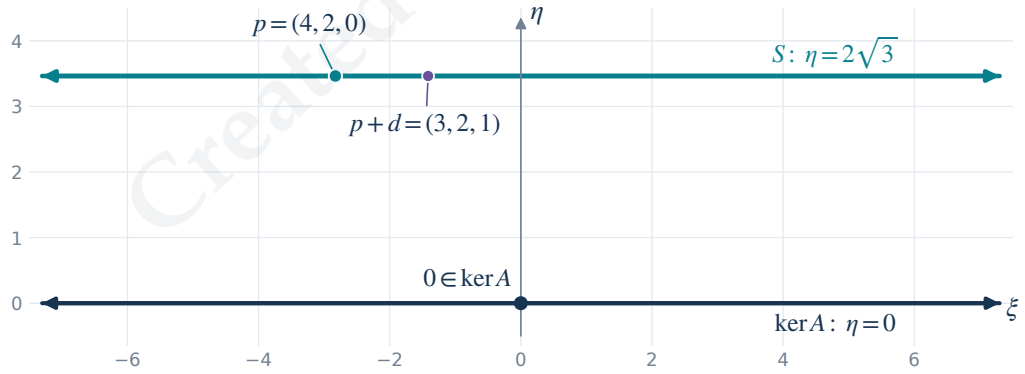
$$(4 - t, 2, t) - (4 - s, 2, s) = (t - s)(-1, 0, 1).$$

Every real multiple occurs by choosing $s = 0$. Thus the differences form exactly $\operatorname{span}\{(-1, 0, 1)\}$, the solution set of the homogeneous system.

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$$S = \{(4-t, 2, t) : t \in \mathbb{R}\} = p + \ker A$$

$$\Phi(\xi, \eta) = \xi u + \eta v, \qquad u = \frac{(-1, 0, 1)}{\sqrt{2}}, \quad v = \frac{(1, 1, 1)}{\sqrt{3}}$$



$$\ker A = \operatorname{span}\{d\}, \quad d = (-1, 0, 1), \qquad 0 \notin S$$

Figure 3.2. The real system $x + y + z = 6$, $x - y + z = 2$ has solution set $S = p + \ker A$, where $p = (4, 2, 0)$ and $\ker A = \operatorname{span}\{(-1, 0, 1)\}$. The ξ, η axes describe the plane $\operatorname{span}\{u, v\} \subseteq \mathbb{R}^3$ through the map Φ displayed in the figure. The vectors u, v are perpendicular unit vectors, so the chart preserves lengths and angles. The translated line excludes the origin, whereas the kernel contains it.

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EXERCISE 3.15 CHALLENGE

Let $a, b \in \mathbb{R}$. Please classify the solutions of

$$\begin{cases} x + y + z = 5, \\ x + ay + z = b, \\ x + y + az = b. \end{cases}$$

In each case, please find the coefficient matrix rank, the number of free variables when consistent, and the solution set. Also find the kernel of the coefficient matrix, and please determine whether the solution set is a subspace.

WORKED SOLUTION

Subtract the first equation from both the second and the third. The equivalent system is

$$\begin{cases} x + y + z = 5, \\ (a-1)y = b - 5, \\ (a-1)z = b - 5. \end{cases}$$

1. If $a \neq 1$, division by $a - 1$ is permitted. Put $c = (b-5)/(a-1)$. The unique solution is

   $$S = \{(5 - 2c, c, c)\}.$$

   There are three pivots, so the rank is 3 and there are no free variables. In the homogeneous system, $y = z = 0$ and then $x = 0$, so $\ker A = \{0\}$.

2. If $a = 1$ and $b = 5$, the last two equations after subtraction are $0 = 0$. The rank is 1, and there are two free variables. Choosing $y = s, z = t$ gives

   $$S = \{(5 - s - t, s, t) : s, t \in \mathbb{R}\} = (5, 0, 0) + \text{span}\{(-1, 1, 0), (-1, 0, 1)\}.$$

   The kernel is $\text{span}\{(-1, 1, 0), (-1, 0, 1)\}$.

3. If $a = 1$ and $b \neq 5$, the equation $0 = b - 5$ is a contradiction, so $S = \varnothing$. The coefficient matrix still has rank 1, while the augmented matrix has rank 2. The kernel depends only on A, so it is the same kernel as in the preceding case.

The solution set is never a subspace: in the consistent cases the zero vector fails to satisfy $x + y + z = 5$, and in the inconsistent case the set is empty. A number of degrees of freedom of a solution set cannot be defined through a choice of solution parameters when no solutions exist.

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EXERCISE 3.16 CHALLENGE

Let $S \subseteq \mathbb{R}^n$ be nonempty and assume $(1-t)x + ty \in S$ for all $x, y \in S$ and $t \in \mathbb{R}$.

1. Please choose $a \in S$ and prove that $U = \{x - a : x \in S\}$ is a subspace and $S = a + U$, where $a + U = \{a + u : u \in U\}$.
2. Is it enough to require only $(x + y)/2 \in S$ for all $x, y \in S$? Please prove or disprove.

WORKED SOLUTION

We have $0 = a - a \in U$. For $u \in U$ and $\lambda \in \mathbb{R}$, the hypothesis gives

$$(1 - \lambda)a + \lambda(a + u) = a + \lambda u \in S,$$

so $\lambda u \in U$. For $u, v \in U$, taking the midpoint of $a + u$ and $a + v$ gives $(u + v)/2 \in U$. Scalar closure with scalar 2 then gives $u + v \in U$. Hence U is a subspace. Its definition gives both inclusions in $S = a + U$. Midpoint closure alone is insufficient. The set $\mathbb{Q} \subseteq \mathbb{R}$ is closed under midpoints. If it were $a + U$ for a real subspace U, the fact that $0 \in \mathbb{Q}$ would imply $\mathbb{Q} = U$. But $1 \in U$ would then imply $\sqrt{3} \cdot 1 \in U$, a contradiction.

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EXERCISE 3.17 CHALLENGE

Consider a finite linear system over a field F with $n \geq 1$ unknowns. Assume every collection of at most $n+1$ of its equations is consistent. Please prove that the full system is consistent. Please construct an inconsistent system of $n+1$ equations for which every collection of n equations is consistent.

WORKED SOLUTION

Suppose the system is inconsistent and choose an inconsistent collection of minimum size k. Write its equations as

$$\sum_{i=1}^{n} a_{ji} x_i = b_j, \qquad 1 \leq j \leq k.$$

The hypothesis gives $k > n+1$. Consider the homogeneous system in $\lambda_1, \ldots, \lambda_k$ given by

$$\sum_{j=1}^{k} \lambda_j a_{ji} = 0 \quad (1 \leq i \leq n), \qquad \sum_{j=1}^{k} \lambda_j b_j = 0.$$

It has $n+1$ equations and more than $n+1$ unknowns. Row reduction therefore gives a nonzero solution. Choose j with $\lambda_j \neq 0$. These identities express equation j, including its right-hand side, as a linear combination of the other equations. Hence every solution of the other equations also satisfies equation j. Minimality of k says that the other equations have a solution. That solution satisfies the whole collection, a contradiction.
The bound is sharp. Take

$$x_1 = 0, \ldots, x_n = 0, \qquad x_1 + \cdots + x_n = 1.$$

The full system is inconsistent. If the final equation is omitted, zero is a solution. If $x_j = 0$ is omitted, setting $x_j = 1$ and all other unknowns to zero gives a solution. Thus every collection of n equations is consistent.

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